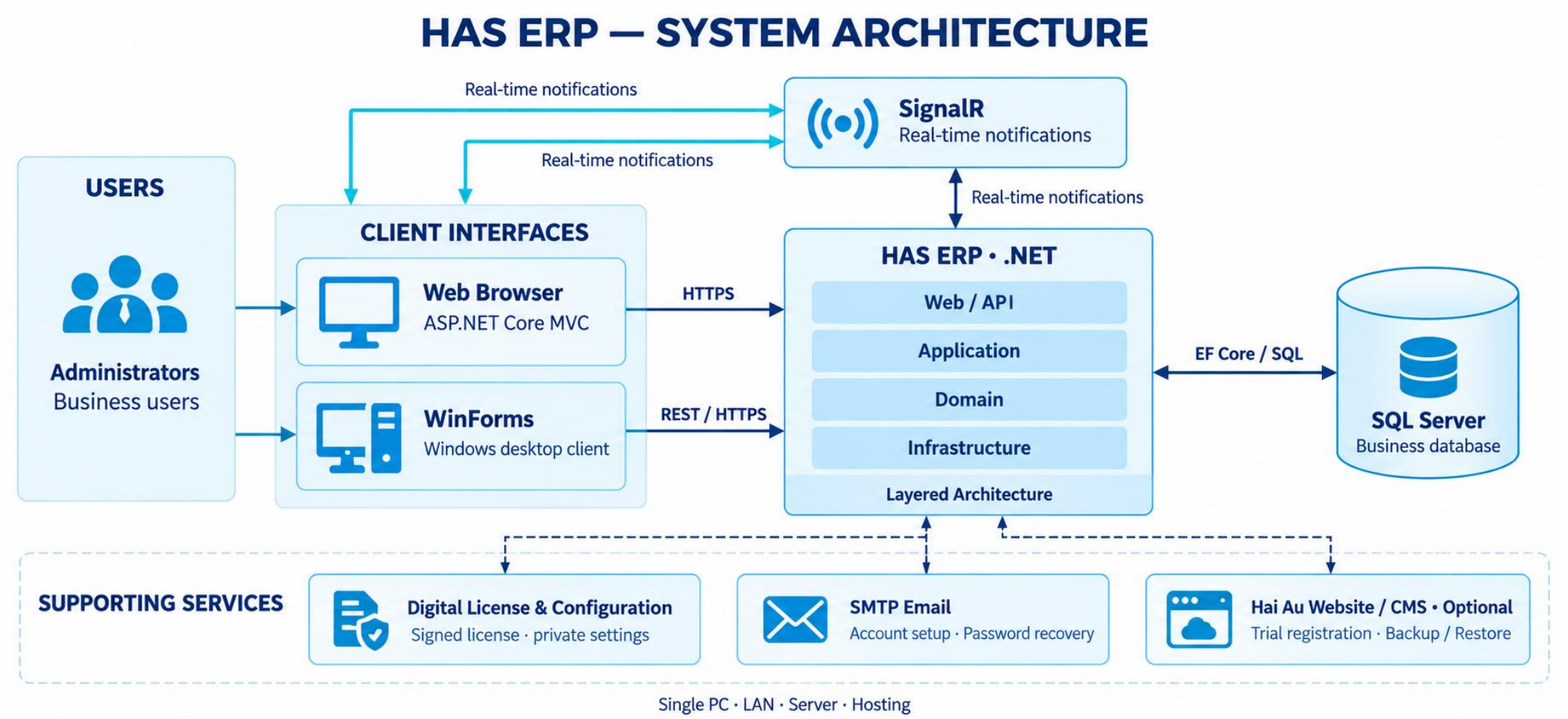
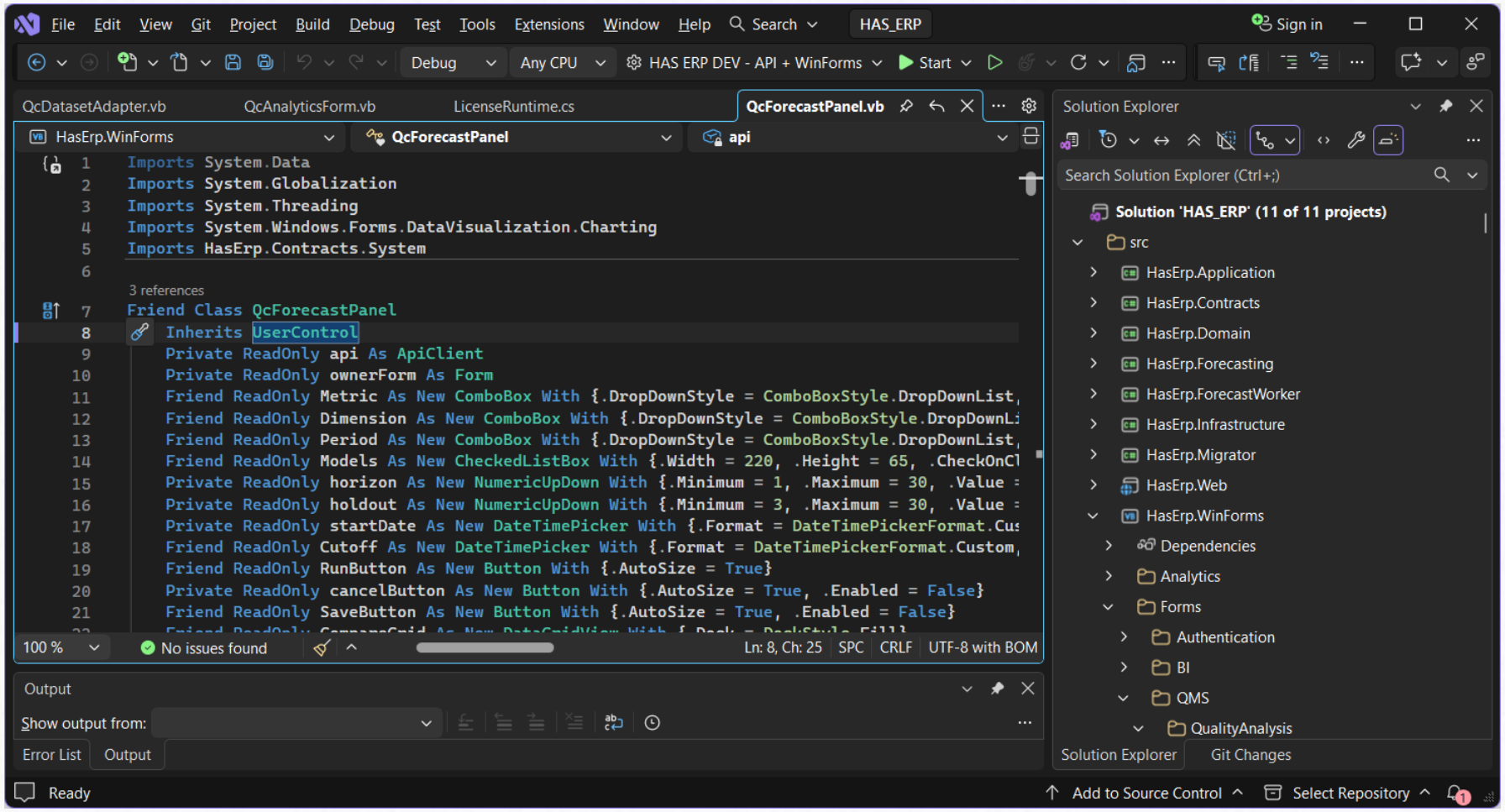
1. Clean Architecture: Systems

HAS ERP combines Web and WinForms interfaces through a shared .NET Web/API platform. Its layered architecture supports centralized SQL Server data, real-time notifications, digital licensing, and email services.



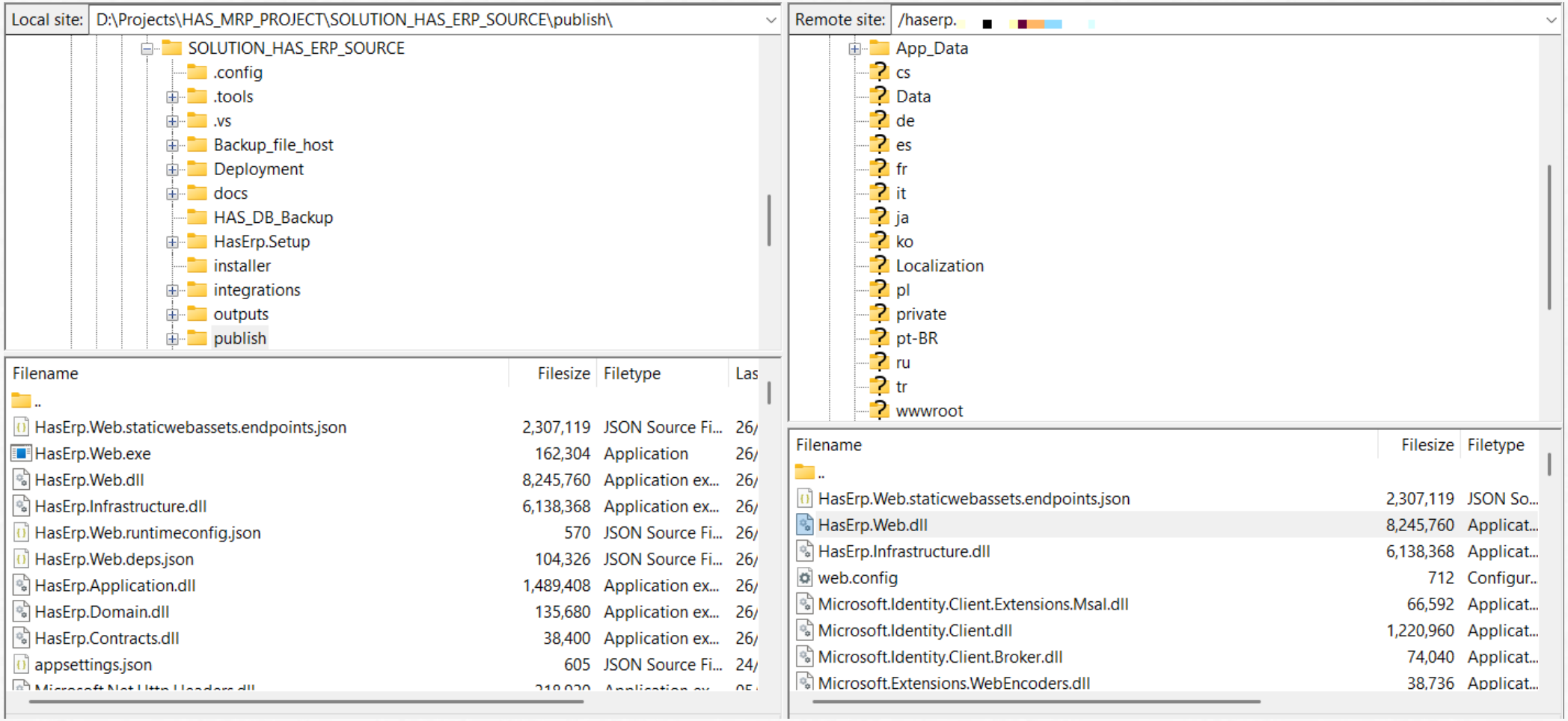
1. Clean Architecture: Code layer

Clean Architecture separates domain logic from application, infrastructure, web, and desktop layers.



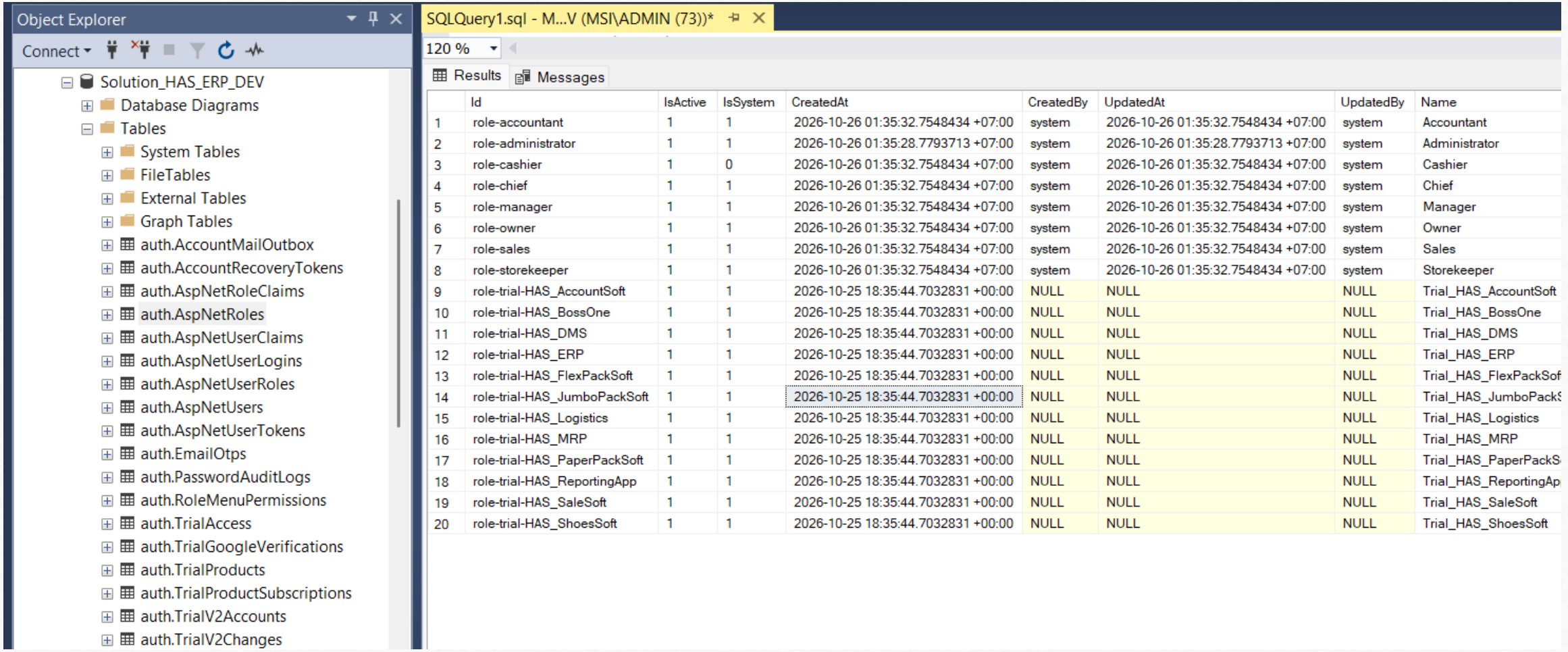
1. Clean Architecture: Hosting deployment

The ERP web application is published and deployed to Windows hosting with its runtime components.



1. Clean Architecture: SQL Server DB

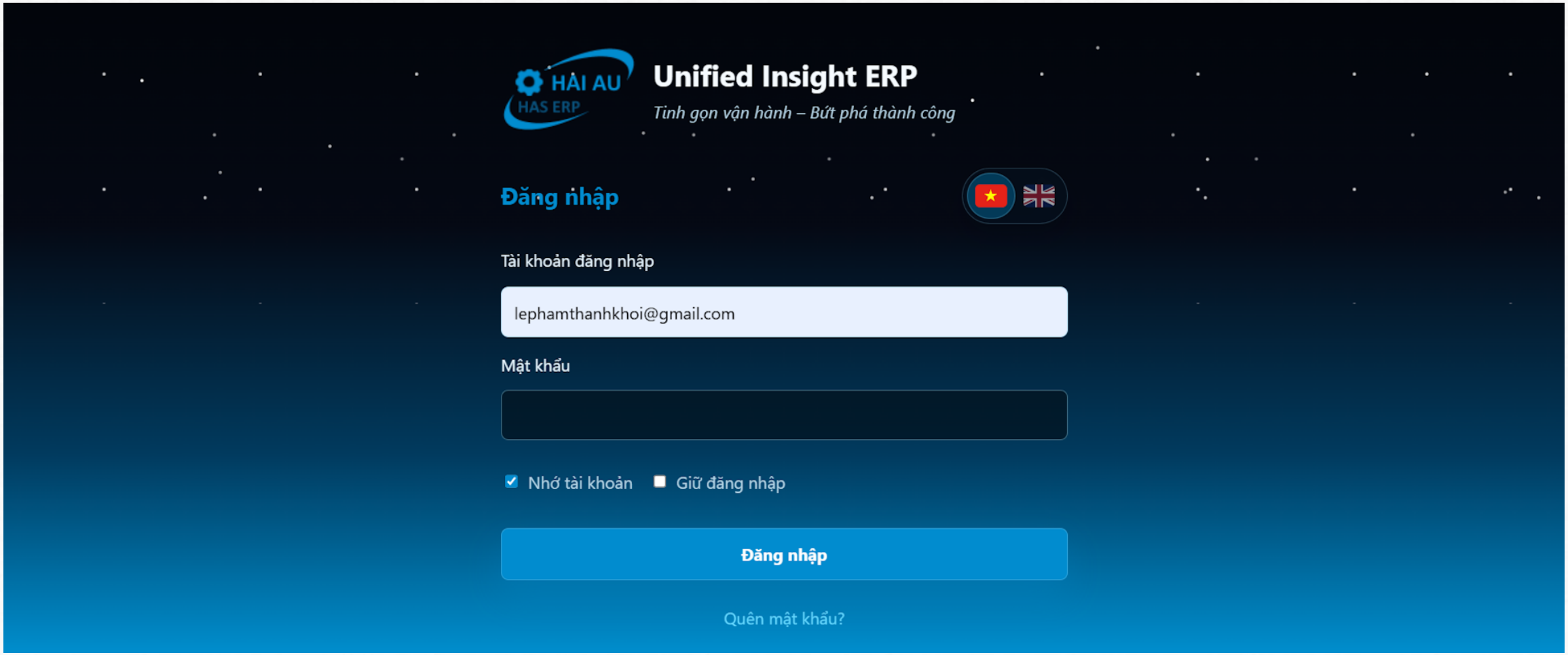
SQL Server centralizes ERP security, master, and transaction data in a structured database.



	Id	IsActive	IsSystem	CreatedAt	CreatedBy	UpdatedAt	UpdatedBy	Name
1	role-accountant	1	1	2026-10-26 01:35:32.7548434 +07:00	system	2026-10-26 01:35:32.7548434 +07:00	system	Accountant
2	role-administrator	1	1	2026-10-26 01:35:28.7793713 +07:00	system	2026-10-26 01:35:28.7793713 +07:00	system	Administrator
3	role-cashier	1	0	2026-10-26 01:35:32.7548434 +07:00	system	2026-10-26 01:35:32.7548434 +07:00	system	Cashier
4	role-chief	1	1	2026-10-26 01:35:32.7548434 +07:00	system	2026-10-26 01:35:32.7548434 +07:00	system	Chief
5	role-manager	1	1	2026-10-26 01:35:32.7548434 +07:00	system	2026-10-26 01:35:32.7548434 +07:00	system	Manager
6	role-owner	1	1	2026-10-26 01:35:32.7548434 +07:00	system	2026-10-26 01:35:32.7548434 +07:00	system	Owner
7	role-sales	1	1	2026-10-26 01:35:32.7548434 +07:00	system	2026-10-26 01:35:32.7548434 +07:00	system	Sales
8	role-storekeeper	1	1	2026-10-26 01:35:32.7548434 +07:00	system	2026-10-26 01:35:32.7548434 +07:00	system	Storekeeper
9	role-trial-HAS_AccountSoft	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_AccountSoft
10	role-trial-HAS_BossOne	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_BossOne
11	role-trial-HAS_DMS	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_DMS
12	role-trial-HAS_ERP	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_ERP
13	role-trial-HAS_FlexPackSoft	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_FlexPackSof
14	role-trial-HAS_JumboPackSoft	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_JumboPack
15	role-trial-HAS_Logistics	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_Logistics
16	role-trial-HAS_MRP	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_MRP
17	role-trial-HAS_PaperPackSoft	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_PaperPackS
18	role-trial-HAS_ReportingApp	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_ReportingAp
19	role-trial-HAS_SaleSoft	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_SaleSoft
20	role-trial-HAS_ShoesSoft	1	1	2026-10-25 18:35:44.7032831 +00:00	NULL	NULL	NULL	Trial_HAS_ShoesSoft

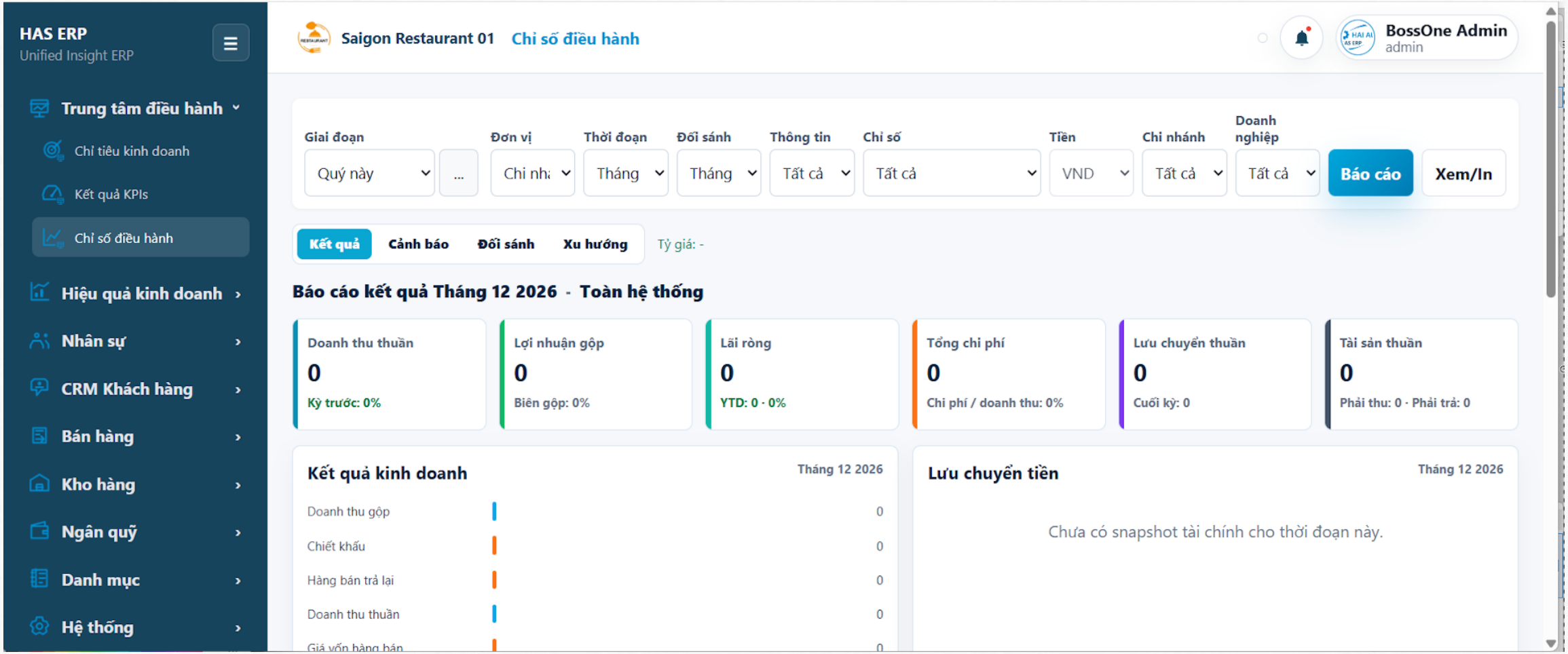
2. Web app: Login

The web login supports secure access, saved credentials, password recovery, and two languages.



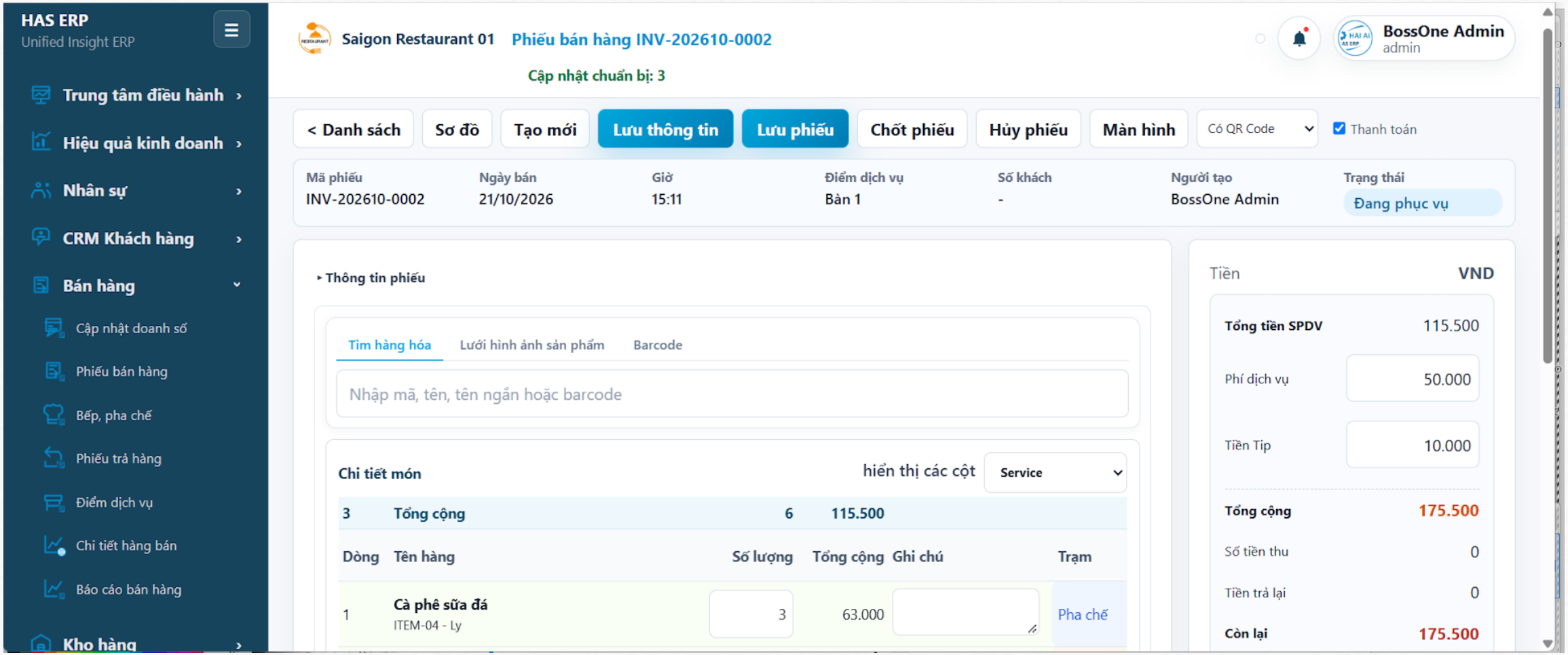
2. Web app: Homepage

The homepage presents business KPIs and role-based navigation for daily operations.



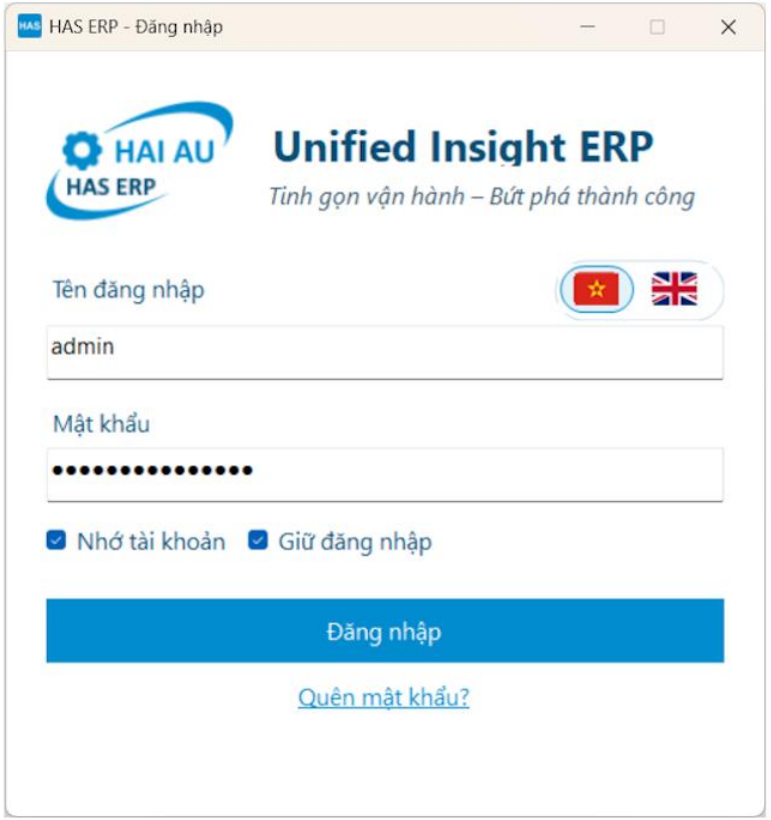
2. Web app: Transaction

The web transaction screen captures sales items, quantities, payments, and order status.



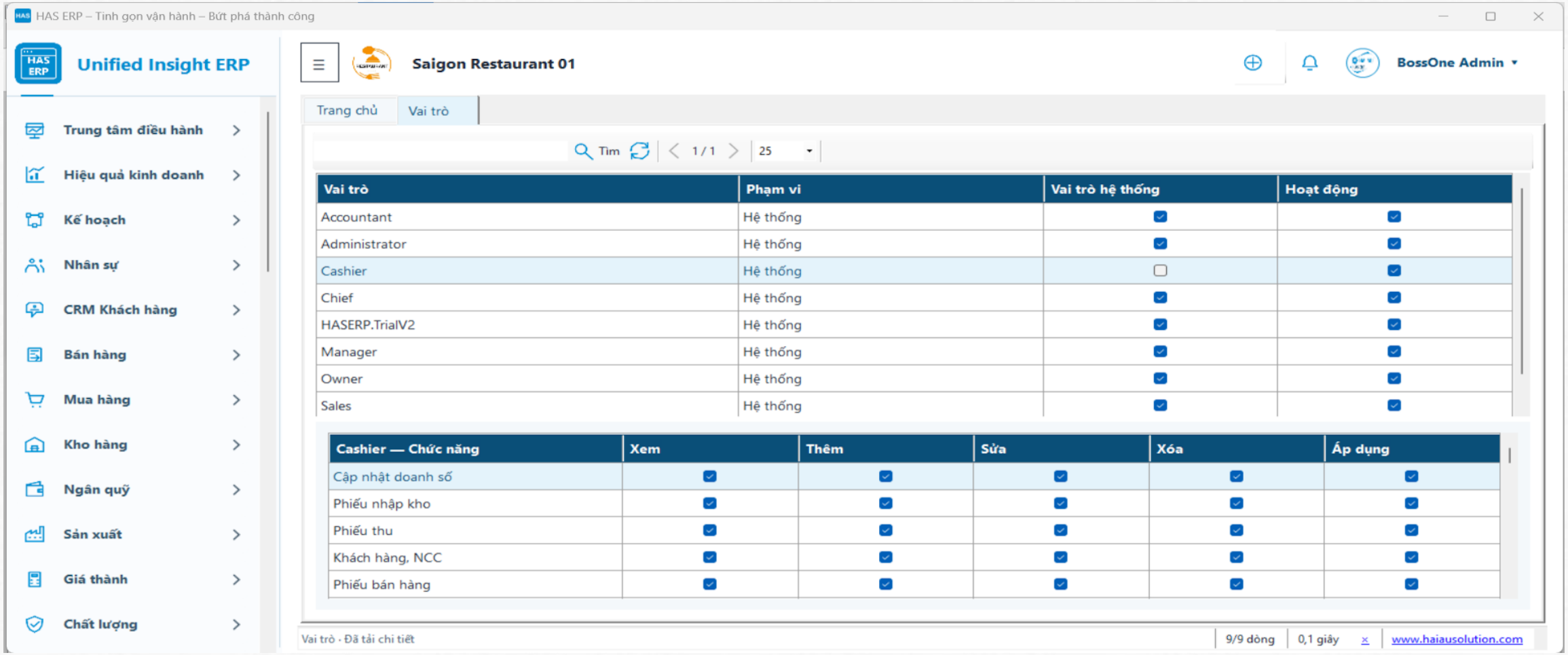
3. Window form: Login

The Windows client supports secure sign-in, saved credentials, password recovery, and two languages.



3. Window form: Master form

The master form manages user roles and detailed permissions across ERP modules.



3. Window form: Transaction

The transaction workspace combines operational data, quality analytics, charts, and detailed records.

